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Filling in the Missing Piece: Advancing Automated Radiology Report Generation with Clinical Insights

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Abstract

*Radiology reports synthesize a radiologist's insights into a patient's condition, serving as vital tools for directing patient care. Despite advances in automated report generation, existing approaches often oversimplify this task, lacking consensus on input selection, ground-truth usage, and output structure. Most methods rely solely on imaging data, merging findings and the impression as a combined output, or incorporate indications as input but generate only the findings. Drawing insights from established medical literature, we clarify the roles and interconnections of report sections and provide definitive answers to some critical questions: (1) What should be the inputs and outputs of a report generation model and why? (2) Can modeling inter-section relationships enhance model performance? We validate our perspective through statistical analyses on MIMIC-CXR, demonstrating universal interdependencies among report sections. We further introduce MIMIC-1V3, a curated dataset derived from MIMIC-CXR where each report is segmented into three sections: the indication, findings, and impression. Furthermore, we propose a **R**adiologist-**L**ike **P**rogressive **G**eneration (RLPG) framework, explicitly modeling relationships among report sections. RLPG mirrors the real-world clinical workflow in two stages: visual understanding to produce factual observations (findings) of the X-ray images, and diagnostic reasoning to interpret the implications of findings (impression), addressing the clinical question in the indication.*

1. Introduction

In real-world medical practice, creating a radiology report (see Fig. 1 (a)) starts with an indication from the ordering physician. The *indication* provides rich clinical context, often containing a clinical question (*i.e.*, the reason for the examination) and the patient's brief medical history. Subsequently, the radiologist interprets the imaging study within the clinical context. The resulting report represents the sum of a radiologist's insight into the patient's condition. It mainly has two sections: findings and impression. The *findings* emphasizes the informative and factual observations of the image, relies on technical language for precision, and provides the basis for the subsequent formulation of the impression. The *impression* addresses the clinical question in the indication and reflects the implications of findings, leading to a diagnosis or differential diagnosis [4].

Creating a clinically acceptable radiology report is time-intensive, requiring years of professional training to master the necessary skills. However, radiologist workload has increased significantly within the last three decades [13]. Automated radiology report generation aims to reduce radiologists' workload by drafting preliminary reports to meet growing diagnostic demands, which has attracted lots of research attention [7, 2, 20, 9, 23, 21, 3]. In retrospect, as shown in Fig. 1, there are two main paradigms in previous works. One paradigm (b-1) simplifies the process by using radiographs as the sole input and combining the findings and impression sections into a single long paragraph as the training target. The other paradigm (b-2) leverages indication and radiographs as inputs but generates only the findings section. Such paradigms have the following limitations: First, combining findings and the impression for training increases the model's learning difficulty due to the distinct nature of these sections. Findings typically involve detailed descriptions of specific structures and abnormalities in the image, while the impression provides a diagnosis that synthesizes these findings within the clinical context [4]. The task of generating both sections simultaneously risks the model disproportionately focusing on the more extended, more detailed findings section at the expense of overlooking the shorter but crucial impression section. As a result, the quality of impression, which require careful diagnostic reasoning, is often compromised.

Second, the impression is closely tied to the indication. For instance, as shown in Figure 1(a), the indication raises a clinical question (*evaluate for volume overload*). Consequently, the impression directly addresses this question

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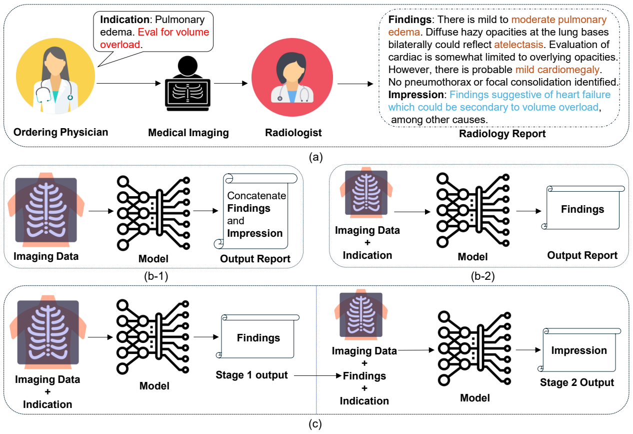

Figure 1. Comparison of real-world radiology workflow (a), two main paradigms in previous works (b-1 & b-2), and our proposed approach (c). The clinical question is highlighted in red. Findings closely related to the indication are highlighted in brown. The impression directly addresses the clinical question (highlighted in blue). Our paradigm takes imaging data and the indication as inputs, progressively generating the findings and impression.

by interpreting the findings (*suggestive of heart failure*) and clearly stating their relevance to the initial clinical concern (*secondary to volume overload*). Hence generating the impression is far beyond summarizing the findings but requires consideration of the patient's medical history and the clinical question. Relying solely on imaging data hinders the model's ability to accurately capture the patterns required for generating the impression. Without the essential context provided by the indication, the model struggles to identify which image details are the most relevant for the diagnosis. This leads to a lack of specificity and accuracy in the impression. This issue becomes particularly problematic when dealing with complex or atypical cases, where the absence of the indication makes it harder for the model to produce clinically meaningful impressions. Besides, omitting the impression section in the output, as seen in the paradigm of Fig. 1 (b-2), deviates from real-world medical practice, leaving the clinical question in the indication unanswered.

As shown in Fig. 1 (c), this work tackles the above limitations by breaking down the report generation process into two consecutive stages: visual understanding and diagnostic reasoning. In the first stage, we only focus on transferring the information from the visual domain to the text domain, *i.e.*, generating the factual observations (the findings) of the given radiograph within the clinical context. In the second stage, we consider what can be inferred from the radiograph that is the most relevant to the clinical question (*i.e.*the impression) aided by the output of the first stage. Finally, we directly combine the findings and impression to yield the final report.

We choose the MIMIC-CXR [8], the largest public available report generation dataset, as our materials. However, the existing statue of MIMIC-CXR is unsuitable for directly validating our proposed paradigm. Specifically, the raw data, including 227,835 free-text reports, exhibits severe inconsistencies. For instance, 57,570 reports lack the findings section, while 69,455 are missing the impression section. Even after applying MIMIC-CXR's official report parsing code, the findings section still contains misassigned contents, such as phrases that belong in the technique or comparison sections. Additionally, some reports include noisy and irrelevant information in the impression section, such as "*Findings were conveyed by Dr. to at 15:33*". These issues create significant obstacles to verifying our approach. To address this, we propose a new benchmark derived from MIMIC-CXR, called MIMIC-1V3. It is a clean, well-structured dataset where each report is segmented into three distinct sections (*i.e.*, indication, findings, impression) and paired with one frontal radiograph. Further details are provided in Section 4. In summary, our contributions include: (1) We propose a new **R**adiologist-**L**ike **P**rogressive **G**eneration (RLPG) framework in the field of report generation, which decomposes the radiologist's workflow into two successive stages: visual understanding and diagnostic reasoning. Our paradigm closely mirrors the real-world medical practice and improves the semantic alignment between input images and output reports, as demonstrated by improved clinical efficacy metrics. (2) Due to no existing large-scale datasets suitable for optimizing our new paradigm, we derive a new benchmark, namely MIMIC-1V3, from the MIMIC-CXR. It serves as a standardized test bed for future report generation models. (3) We conduct quantitative evaluations to assess the impact of modeling the interdependence of report sections on the quality of the generated findings and impression. Additionally, we compare the performance of our approach against advanced LLM-based report generation models and demonstrate that it significantly outperforms them.

2. Related Works

The usage of ground truth data of previous efforts in report generation diverges. Some works [7, 2, 20] combine the findings and impression sections as the training target and take images as the sole input. In the groundbreaking work of [7], the authors explain this practice by stating: "The impression and findings sections are concatenated together as a long paragraph since impression can be viewed as a conclusion or topic sentence of the report." Other studies [10, 17] focus on generating the findings of given radiographs. Recent works [5, 1] have recognized the rich clinical context in the indication section, thereby using indication and radiographs as model inputs while generating only the findings section. The exclusion of the impression section in these works is either unspecified or justified by the assertion that "the impression section summarizing the actionable insights from the study... cannot be fully gathered from the image alone."

Advancements in report generation have also seen the incorporation of large-scale training datasets and the adoption of instruction tuning techniques to build large language model (LLM)-centered multimodal multitask interactive systems. For example, RadFM [21], CheXagent[3], and MedVersa [23] scale up the training data to the millions level, covering nearly all available public medical datasets. In their settings, the task of report generation reduces to a downstream task. CheXagent divides report generation into two tasks: (1) generating findings from the image and (2) generating impression directly from the image or through findings summarization without the image. MedVersa designs different prompts for generating findings-only, impression-only, and complete report (findings + impression). RadFM designs the prompt for report generation as "Please generate a radiology report for this scan *<image-1>*?", without specifying which sections to generate. Other notable works such as LLM-CXR [9] and R2GenGPT [20], both obfuscate findings and impression sections, instructing the model to generate a "free-text radiology reports" or "a comprehensive and detailed diagnosis report" for the given radiograph. These works overlook the importance of the indication and fail to leverage it to guide the generation of the findings and impression.

3. Method

The overview of **R**adiologist-**L**ike **P**rogressive **G**eneration (RLPG) framework is illustrated in Fig. 2. It consists of two consecutive stages, *i.e.*, findings generation and impression generation. In the first stage, the image encoder extracts the image feature, which are then projected into the latent space of the LLM via the projection layer as image embeddings. We concatenate them with text embeddings of the indication and an instruction prompt. The LLM processes these integrated embeddings to yield detailed findings. In the second stage, the image embeddings are concatenated with text embeddings of the same indication, a new instruction prompt, and the findings as the inputs of the LLM. The LLM only generates the impression in the second stage. We use ground-truth findings during training and predicted findings during inference for the impression generation. We depict more details in the following.

3.1. Problem Reformulation

In a typical radiology report generation model, the objective is to maximize the probability $p(y|x)$ between input image x and output report y, which is a non-trivial task due to the intrinsic complicity in medical data and the large modality gap between visual inputs and textual outputs. To bridge the modality gap and tackle the intrinsic complicity

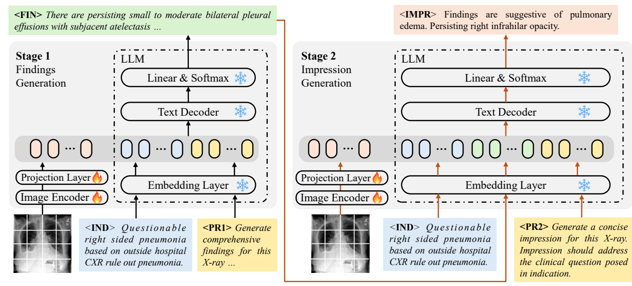

Figure 2. Overall framework. In the first stage, the inputs are image embeddings and text embeddings sourced from the indication (IND) and instruction prompt (PR1). The output is the findings (FIN) section. In the second stage, the inputs are image embeddings combined with text embeddings of the same indication, a new instruction prompt (PR2), and the FIN. The output is the impression (IMPR) section.

in medical data, we draw inspiration from the workflow of radiologists and decompose the complex process of radiology report generation into two sequential stages and conquer them progressively. Generally, a normative radiology report y consists of a findings u and an impression v, *i.e.*, $y \leftarrow [u, v]$. In the first stage, the model takes image x as input and generates only findings u. In the second stage, the models takes the image x and the generated findings u as input and outputs the impression v. Mathematically, the probability becomes

$$p(y|x) = p(u, v|x) = p(v|x, u)p(u|x). \tag{1}$$

Moreover, to further reduce the risk of overlooking critical clinical information and enhance diagnostic accuracy, we mimic the process of radiologists interpreting the radiographs in regard to the indication. Concretely, we introduce the indication as input for both stages mentioned above. The indication directs the model to focus more intently on specific anatomical locations or abnormalities within the image x, ensuring a more thorough analysis. Thus, we reformulate Eq. (1) to

$$p(y|x, c) = p(u, v|x, c) = p(v|x, c, u)p(u|x, c), \tag{2}$$

where c refers to the indication corresponding to x.

3.2. Radiologist-Like Progressive Generation (RLPG) Framework

Stage 1: Findings Generation As shown in Fig. 2 (left), the process begins with encoding the radiograph x into the visual embeddings via the image encoder $\mathcal{E}_{img}$ and the projection layer $\mathcal{P}^{(1)}$. Concurrently, textual inputs—including the indication c that provides crucial clinical context, and the instruction prompt r_1 —are transformed into textual embeddings by the embedding layer $\mathcal{E}_{txt}$ of the LLM. The visual and textual embeddings are concatenated then fed through subsequent layers $\mathcal{D}_{txt}$ of the LLM, which processes the data to synthesize detailed and clinically relevant findings u. Mathematically, the findings generation process $\mathcal{G}^{(1)}$ can be defined as

$$u = \mathcal{G}^{(1)}(x, c, r_1) = \mathcal{D}_{txt}\left(\left[\mathcal{P}^{(1)}\left(\mathcal{E}_{img}(x)\right), \mathcal{E}_{txt}([c, r_1])\right]\right),$$

where $[\cdot, \cdot]$ refers to the concatenation operation. These findings encapsulate the critical observations derived from the image, which is useful for clinical decision-making.

Stage 2: Impression Generation As shown in Fig. 2 (right), the same image x is re-encoded via the image encoder $\mathcal{E}_{img}$ and a projection layer $\mathcal{P}^{(2)}$. These visual embeddings are concatenated with the textual embeddings of the same indication c, a new instructional prompt r_2, and the findings u. This comprehensive input setup ensures that the impression generation is informed by both the findings and additional context that may influence the clinical interpretation. Similar to the first stage, the input then is processed by the rest of the LLM, *i.e.*, $\mathcal{D}_{txt}$. Formally, the impression generation $\mathcal{G}^{(2)}$ can be written as

$$v = \mathcal{G}^{(2)}(x, c, u, r_2) = \mathcal{D}_{txt}\left(\left[\mathcal{P}^{(2)}(\mathcal{E}_{img}(x)), \mathcal{E}_{txt}([c, u, r_2])\right]\right),$$

where $\mathcal{E}_{txt}$ is the embedding layer of the LLM, and v is the generated impression. This impression provides the necessary contextualization and diagnosis to guide further medical action or evaluation.

3.3. Training and Inference

Training Based on the formulation above, our training objective can be defined as

$$\begin{aligned}\max_{\mathcal{E}_{img}, \mathcal{P}} \log p(y|x, c) &= \max_{\mathcal{E}_{img}, \mathcal{P}} \log p(u, v|x, c) \\ &= \max_{\mathcal{E}_{img}, \mathcal{P}} \underbrace{\log p(v|x, c, u)}_{\text{second stage}} + \underbrace{\log p(u|x, c)}_{\text{first stage}}.\end{aligned}$$

Here, we only optimize the image encoder $\mathcal{E}_{img}$ and projection layer $\mathcal{P}$ while keeping the LLM parameters fixed. This way refines visual feature extraction for our specific needs without disturbing the established linguistic capabilities of the LLM, ensuring stable text generation. Notably, we also omit the instruction prompts r_1 and r_2 since they are consistent across all samples during both the training and inference phases. In the first stage, we initialize the image encoder $\mathcal{E}_{img}$ with ImageNet pre-trained weights and randomly initialize the projection layer $\mathcal{P}$. In general, sequence generation models are often trained using the autoregressive Teacher Forcing technique, which maximizes the likelihood of the often token w_t given all previous often tokens $w_{i<t}$. In our setting, we optimize the model by

$$\mathcal{L}_1 = \sum_{t=1}^{T} -\log p(w_t|w_{i<t}, x, c), \tag{3}$$

where w_t is the t-th token in the findings u and T is the total number of tokens in u.

We take the well-trained image encoder in the first stage as an initialization for the second-stage image encoder with a new randomly initialized projection layer. Similar to the optimization method in stage one, the loss function in the second stage can be written as

$$\mathcal{L}_2 = \sum_{l=1}^{L} -\log p(w_l|w_{j<l}, x, c, u), \tag{4}$$

where w_l is the l-th token in the impression v and L is the total number of tokens in v. We use ground-truth findings during training and the predicted findings in inference.

Inference In the first stage, with the pre-defined instruction prompt r_1, the whole model $\mathcal{G}^{(1)}$ takes a radiograph x and the indication c as inputs to generate findings $\tilde{u}$. In the second stage, with another prompt r_2, both the generated findings $\tilde{u}$ and the same inputs x and c are fed into model $\mathcal{G}^{(2)}$ to produce an impression $\tilde{v}$. We acquire the final report $\tilde{y}$ by combining two outputs, *i.e.*, $\tilde{y} \leftarrow [\tilde{u}, \tilde{v}]$. Formally, the whole inference process can be defined as

$$\tilde{y} \leftarrow [\tilde{u}, \tilde{v}], \text{ where } \tilde{u} = \mathcal{G}^{(1)}(x, c, r_1) \text{ and } \tilde{v} = \mathcal{G}^{(2)}(x, c, \tilde{u}, r_2).$$

This structured, sequential approach allows each model to specialize in distinct aspects of report generation, enhancing the overall accuracy and relevance of the generated reports.

4. MIMIC-1V3 Dataset Construction

Raw reports from MIMIC-CXR are unstructured with varying numbers of sections. For example, 12.5% of all reports do not have the findings section. We aim to provide a clean and structured dataset with easy access to different report sections. The construction of MIMIC-1V3 mainly comprises three steps:

View Selection The MIMIC-CXR dataset contains 14 unique X-ray image views. The frontal view, comprising Anterior-Posterior (AP) and Posterior-Anterior (PA) views, makes up 64.5% of all images. The number of images paired with each report varies, with 45.4% of reports associated with just one image. We exclusively pair one frontal view image with each report for consistency.

Expanding Abbreviations & Acronyms Abbreviations and acronyms are frequently used in the indication section of radiology reports, often leading to ambiguity and reduced clarity. To address this, we curate a standardized list of medical abbreviations from Radiopaedia[1] and expand them into their full textual forms. For example, the raw indication "M with c/o left lower CP with SOB and cough. ? PNA." is inherently ambiguous. We restore it to "Male with complaints

[1] https://radiopaedia.org/

of chest pain with shortness of breath and cough. ?pneumonia," thereby enhancing readability and semantic clarity.

Report Parsing & Cleaning We use MIMIC-CXR's official code base to parse free text reports, extracting the indication, findings, and impression sections while excluding the incorrectly categorized *comparison* and *technique* sections. For example, phrases like "*Two frontal chest radiographs were obtained with patient positioned upright*" are placed under findings instead of technique. To provide a cleaner version of the report, we reviewed approximately 20,000 reports to identify patterned phrases and then automatically remove such misassigned phrases from other reports. Additionally, we discard reports lacking all three sections, for example, 59,628 reports contain only indication and impression, leaving it unclear whether the findings section is absent or misassigned during data collection.

Additionally, in the supplementary materials, we provide detailed dataset analysis including the data split and label distribution of MIMIC-1V3 in Section 1 and case studies in Section 2 which showcase the interconnections among different report sections and the necessities of including the indication as the input.

5. Experimental restuls

5.1. Implementations and Metrics

Implementation details. During training, we employ Swin Transformer [11]-base as the image encoder for both stages, one linear layer as the projection layer, and Llama2-7B-chat [18] as our LLM. The image encoders and projection layers are trainable, while the LLM remains frozen and shared across stages. For stage 1, we instruct the LLM to generate findings conditioned on image and indication and explicitly define the findings as the factual observation of the X-ray image. For stage 2, we include the findings as part of the prompt. The LLM is instructed to generate the impression based on the image, indication, and findings, focusing on addressing the clinical question in the indication. We use AdamW [12] as the optimizer and cosine scheduler with a learning rate of 3e-5 and a weight decay of 0.01. The majority of our experiments are conducted on two NVIDIA L40S GPUs. During inference, the model first generates *findings* based on image and indication. We use generated *findings*, image, and indication to generate the impression.

Evaluation metrics. We use BLEU-4 (B-4) [15] and CIDEr [19] as Natural Language Generation (NLG) metrics. We employ Bert-Score (Bert-S) [22], Chexbert-all Micro F1 (F1-all) [16], and RadGraph-related metrics [6]: RadEntity-ExactMatch (RE-ME), RadEntity-NLI (RE-NLI) [14], and RadGraph-Complete (Rad-C) as Clinical Efficacy (CE) metrics.

5.2. Analysis of Evaluation Results

We analyze evaluation results on both validation and test sets. Results in Table 1 and Table 2 are obtained with the same model architecture. The differences among the settings are the input data, output sections, and training stages (*i.e.*, one stage or progressive two stages).

Advantages of Our Progressive Training In Table 1, we provide evaluation results of baseline (one-stage training) and our proposed progressive paradigm. In the baseline setting, the model receives only the image as input and simultaneously generates both the findings and impression sections. In contrast, our paradigm progressively generates the findings then the impression section. We separately evaluate each section. On the validation set, the baseline model yields a CIDEr score of 0.3084 and an F1-all score of 0.4041 for the findings section. However, it yields lower scores for the impression section, with a CIDEr score of 0.2726 and a Bert-Score of 0.3518, indicating that the baseline model exhibits greater difficulty in generating the impression section. In contrast, our proposed progressive paradigm demonstrates advantages over the baseline, especially in generating the impression section. On the val split, the CIDEr score of the findings increases from 0.3084 to 0.4293 and the F1-all score increases from 0.4041 to 0.4617. More notably, for the impression section, when using predicted findings as input, the CIDEr score increases from 0.2726 to 1.1259, and the Bert-Score increases from 0.3518 to 0.4787. Elevated evaluation scores prove our progressive generation strategy can enhance clinical efficacy and text generation quality.

Despite the label frequency and section length shift between the training and test sets, which causes a noticeable performance discrepancy between validation and test sets, we observe the advantages of our progressive paradigm on the test set. Specifically, with the progressive paradigm, the CIDEr score of the findings section increases from 0.1530 to 0.1663. For the impression section, the CIDEr score improves from 0.4530 to 0.6060, and the F1-all score rises from 0.3875 to 0.4301. These results highlight that, despite the distributional shifts between splits, the progressive paradigm demonstrates stronger generalization ability, particularly in generating clinically meaningful impressions. In addition, on validation and test sets, replacing the predicted findings with ground-truth findings for impression generation results in a considerable leap across all metrics. This denotes that the quality of the findings dramatically impacts the impression quality. Overall, even though the distributional shift affects the test set performance, the progressive approach consistently shows its superiority in generating higher-quality findings and impressions.

Benefits of Using Indication as Input In this part, we explore the benefits of including the indication as input in our progressive framework. To this end, we compare the

Table 1. Model performance of baseline one-stage training and our progressive paradigm. Here, we abbreviate image, indication, findings, and impression as IMG, IND, FIN, and IMPR. Symbols indicate the following: ✓= using images or ground truth findings as input, ✓= using predicted findings as input, and ✓= sections that are model outputs.

Dataset	Split	Input			Output		Findings							Impression						
							NLG		Clinical Efficacy					NLG		Clinical Efficacy				
		IMG	IND	FIN	FIN	IMPR	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑
MIMIC-1V3	val	✓			✓	✓	0.1169	0.3084	0.5744	0.4041	0.3884	0.3238	0.2285	0.0429	0.2726	0.3518	0.5025	0.2310	0.2803	0.1187
		✓			✓		0.1588	0.4293	0.5766	0.4617	0.4095	0.3619	0.2425	-	-	-	-	-	-	-
		✓		✓		✓	-	-	-	-	-	-	-	0.0824	1.1259	0.4787	0.5057	0.3017	0.3403	0.2011
		✓		✓		✓	-	-	-	-	-	-	-	0.2430	2.9432	0.6342	0.7125	0.5439	0.5246	0.4139
	test	✓			✓	✓	0.1044	0.1530	0.5595	0.4476	0.3631	0.2363	0.1981	0.0532	0.4530	0.3277	0.3875	0.1539	0.1657	0.0869
		✓			✓		0.1139	0.1663	0.5437	0.4225	0.3320	0.2360	0.1873	-	-	-	-	-	-	-
		✓		✓		✓	-	-	-	-	-	-	-	0.0582	0.6060	0.4411	0.4301	0.2374	0.1686	0.1481
		✓		✓		✓	-	-	-	-	-	-	-	0.1948	1.5391	0.5715	0.6475	0.4504	0.2853	0.3081

Table 2. Evaluation results when including the indication as input. In stage one, the model generates findings from the image and indication. In stage two, it generates the impression using the image, indication, and predicted findings.

Dataset	Split	Input			Output		Findings							Impression						
							NLG		Clinical Efficacy					NLG		Clinical Efficacy				
		IMG	IND	FIN	FIN	IMPR	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑
MIMIC-1V3	val	✓			✓		0.1588	0.4293	0.5766	0.4617	0.4095	0.3619	0.2425	-	-	-	-	-	-	-
		✓	✓		✓		0.1825	0.6376	0.5906	0.4723	0.4192	0.3569	0.2554	-	-	-	-	-	-	-
		✓		✓		✓	-	-	-	-	-	-	-	0.0824	1.1259	0.4787	0.5057	0.3017	0.3403	0.2011
		✓	✓	✓		✓	-	-	-	-	-	-	-	0.1075	1.6675	0.5239	0.5475	0.3647	0.4159	0.2499
		✓		✓		✓	-	-	-	-	-	-	-	0.2430	2.9432	0.6342	0.7125	0.5439	0.5246	0.4139
		✓	✓	✓		✓	-	-	-	-	-	-	-	0.2705	3.1813	0.6485	0.7133	0.5569	0.5830	0.4252
	test	✓			✓		0.1139	0.1663	0.5437	0.4225	0.3320	0.2360	0.1873	-	-	-	-	-	-	-
		✓	✓		✓		0.1155	0.2856	0.5630	0.4516	0.3685	0.2430	0.2026	-	-	-	-	-	-	-
		✓		✓		✓	-	-	-	-	-	-	-	0.0582	0.6060	0.4411	0.4301	0.2374	0.1686	0.1481
		✓	✓	✓		✓	-	-	-	-	-	-	-	0.0648	0.7547	0.4581	0.4308	0.2773	0.2206	0.1727
		✓		✓		✓	-	-	-	-	-	-	-	0.1948	1.5391	0.5715	0.6475	0.4504	0.2853	0.3081
		✓	✓	✓		✓	-	-	-	-	-	-	-	0.2074	1.6740	0.5897	0.6614	0.4892	0.3513	0.3306

performance of our framework with and without the indication as input. As shown in Table 2, on the validation set, after incorporating the indication as inputs, the BLEU-4 and CIDEr scores of the findings increase from 0.1588 to 0.1825 and from 0.4293 to 0.6376, respectively. All CE scores have improved except the RadEntity-NLI.

The benefit of using the indication as input is more evident on generating impression than findings. The improvement can be observed across all evaluation metrics, such as CIDEr score increases from 1.1259 to 1.6675 and RadEntity-NLI score increase from 0.3403 to 0.4159. After replacing the predicted findings with ground-truth findings for impression generation, the indication can still further enhance the quality of the impression, improving BLEU-4 score from 0.2430 to 0.2705, CIDEr score from 2.9432 to 3.1813, and RadEntity-NLI score from 0.5246 to 0.5830.

On the test set, despite performance drops compared to the validation set due to label and length shifts, incorporating the indication as input leads to notable improvements in findings generation. Specifically, the CIDEr score increases from 0.1663 to 0.2856, the RadEntity-ExactMatch score rises from 0.3320 to 0.3685, and the RadGraph-Complete score improves to 0.2026. For impression generation, incorporating the indication results in an increase in the BLEU-4 score from 0.0582 to 0.0648, the CIDEr score from 0.6060 to 0.7547, the RadEntity-NLI score from 0.1686 to 0.2206, and the RadGraph-Complete score rises to 0.1727.

These improvements are due to the indication serving as a guideline for the generation of findings, providing context and a soft boundary for what should be considered in the findings that are closely related to the patient's medical history and the reason for the exam. The indication and impression are similar to a question-answer pair where the indication raises the question, and the impression addresses it. Using the indication as the input could provide an efficient supervision signal for impression generation.

Comparison with Other Models We evaluate five representative LLM-based models, *i.e.*R2GenGPT [20], RadFM [21], CheXagent [3], LLM-CXR [9] and MedVersa [23], on the MIMIC-1V3 test set. RadFM is highly sensitive to prompts for report generation. The authors provide 35 prompts (15 caption prompts and 20 report prompts), but we are unable to identify the optimal prompt for the task. Among the prompts we experiment with, "*Can you provide a radiology report for this medical image?*" yields the best evaluation results. For CheXagent, we follow the prompts specified in its original paper: (1) Given the <*image*>, generate its <*findings*>; (2) Given the <*image*>, generate its <*impression*>; and (3) Given the <*findings*>, generate its <*impression*>.

Table 3. Comparison with other LLM-based RRG methods.

Dataset	Method	Input			Output		Findings							Impression						
		IMG	IND	FIN	FIN	IMPR	NLG		Clinical Efficacy					NLG		Clinical Efficacy				
							B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑	B-4 ↑	CIDEr ↑	Bert-S ↑	F1-all ↑	RE-EM ↑	RE-NLI ↑	Rad-C ↑
MIMIC-1V3	R2GenGPT	✓			✓	✓	0.1044	0.1530	0.5595	0.4476	0.3631	0.2363	0.1981	0.0532	0.4530	0.3277	0.3875	0.1539	0.1657	0.0869
	RadFM	✓			✓	✓	0.0236	0.0576	0.3917	0.2060	0.2381	0.1846	0.1224	0.0164	0.5072	0.2289	0.2802	0.1344	0.1431	0.0893
	CheXagent	✓			✓		0.0540	0.0732	0.5152	0.2588	0.3484	0.2639	0.1740	-	-	-	-	-	-	-
	CheXagent	✓				✓	-	-	-	-	-	-	-	0.0305	0.6147	0.3807	0.3749	0.2169	0.1753	0.1323
	CheXagent			✓		✓	-	-	-	-	-	-	-	0.0229	0.4732	0.3881	0.3075	0.1496	0.1624	0.0928
	LLM-CXR	✓			✓	✓	0.0032	0.0072	0.3344	0.3028	0.1308	0.0294	0.0514	0.0208	0.2608	0.3593	0.3502	0.1803	0.0946	0.0860
	MedVersa	✓			✓	✓	0.0003	0.0066	0.2492	0.3615	0.1362	0.0401	0.0581	0.0025	0.0477	0.3191	**0.4420**	0.1789	0.0113	0.0690
	Ours	✓	✓		✓		**0.1155**	**0.2856**	**0.5630**	**0.4516**	**0.3685**	**0.2430**	**0.2026**	-	-	-	-	-	-	-
	Ours	✓	✓	✓		✓	-	-	-	-	-	-	-	**0.0648**	**0.7547**	**0.4581**	0.4308	**0.2773**	**0.2206**	**0.1727**

As shown in Table 3, our framework outperforms other LLM-based models by a large margin. RadFM shows limited effectiveness in generating clinically meaningful reports, achieving the lowest BLEU-4 scores for both findings and impression among all models. Although CheXagent achieves improved evaluation results in the findings section, it remains inferior to our framework, obtaining BLEU-4 scores of 0.0540 versus 0.1155, CIDEr scores of 0.0732 versus 0.2856, and F1-all scores of 0.2588 versus 0.4516. Conversely, for CheXagent, generating the impression directly from the image yields better results than generating the impression by summarizing its own generated findings. This observation suggests that the findings produced by CheXagent are less informative than the original image data. Consequently, incorporating the image as input for impression generation proves beneficial, particularly when the quality of the generated findings is sub-optimal. The innovation of LLM-CXR lies in its capacity to perform bidirectional generation between medical images and textual reports. However, our evaluation reveals instances of factual inconsistencies (hallucinations) in its generated outputs, which is reflected in relatively lower performance on standard NLG and clinical efficacy (CE) metrics. Additionally, MedVersa provides limited documentation regarding large-scale inference procedures and the hyperparameter configurations required for optimal generation. These factors currently hinder its reproducibility and practical deployment.

While RadFM and CheXagent can perform multiple tasks and handle modalities beyond X-rays, R2GenGPT is exclusively dedicated to chest X-ray report generation. Consequently, it is unsurprising that R2GenGPT's performance is comparable to our proposed approach. It is important to note that RadFM leverages a domain-specific LLM with 13 billion parameters, while CheXagent fine-tunes a Mistral-7B model on their medical corpus. In contrast, our method employs a general Llama2-7B-Chat model without domain-specific pretraining. By incorporating the indication as input and adopting a progressive generation strategy, our approach effectively compensates for the lack of domain-specific knowledge within the base LLM.

In summary, we demonstrate the quantitative impact of integrating indication as inputs on the quality of generated findings and impression and conclude that: 1) our radiologist-like progressive generation paradigm *i.e.*, generating findings and the impression, is more effective than generating both sections at once, and 2) incorporating indication as part of input can benefit the generation of findings and impression on both NLG and CE metrics.

In addition, we visualize generated reports in section 3 of the supplementary materials to further demonstrate the advantages of our proposed paradigm. In section 4.1, we presents an ablation study on the impact of incorporating images for impression generation, showing that visual input can compensate for missing or inaccurate information in the findings and enhance model robustness. In section 4.2, we further provide evaluation results of R2GenGPT, RadFM, CheXagent, LLM-CXR and MedVersa with the indication as input. As their models do not take the indication as input during their extensive pretraining, the effect of adding the indication is inconclusive.

6. Conclusion and Future Works

This work proposes a novel radiologist-like progressive generation (RLPG) framework for automated radiology report generation, consisting of two successive stages: visual understanding and diagnostic reasoning. RLPG explicitly leverage the interconnections among different report sections, closely mimicking real-world radiology workflows and resulting in more clinically accurate reports than other LLM-based report generation models. Furthermore, we introduce MIMIC-1V3, a structured benchmark derived from MIMIC-CXR, where each report is segmented into indication, findings, and impression and paired with a frontal view radiograph. Compared to MIMIC-CXR, MIMIC-1V3 is cleaner and more consistent across all samples.

Future research could expand the framework and dataset to address different radiological sub-specialties, enhance natural language understanding for better clinical language interpretation, and incorporate temporal analysis to track patient condition changes over time. Testing the framework's adaptability across diverse datasets, especially from different anatomical regions or healthcare systems, will ensure its generalizability and robustness. Moreover, enriching the MIMIC-1v3 dataset with more detailed annotations, such as disease severity, could increase the utility and clinical relevance of the automated reports.

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